Machine Learning in Production
Midterm 2, Fall 2025

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Instructions:

- Including this cover sheet and the scenario, your exam should have **7** pages. Make sure you are not missing any pages. *You may detach the last page and recycle it after the exam.*
- All questions in this midterm refer to the scenario on **Page 7**. Answers are graded in the context of the scenario; **generic answers that do not relate to the scenario will not receive full credit.**
- The exam has a maximum score of **50** points. The point value of each problem is indicated. We designed the exam anticipating approximately one minute per point, but you have 80 minutes to complete it.
- **Please write legibly**. We are unlikely to be able to grade your solution if we can't read it.
- We give an amount of space commensurate with what we expect you to need for each question. We use horizontal lines to suggest where to not use the full page. You may exceed those limits if it is clear where to find the rest of your answer. However, we strongly recommend writing concise, careful answers; short and specific is much better than long, vague, or rambling. However, **do NOT write anything you want us to grade on the back of pages.** We will scan the exam and will not look at the back sides.
- This is a **closed book exam**; no books or electronics allowed. You may refer to 6 sheets of notes (handwritten or typed, both sides).

Question 1: MLOps [18 points]

All questions in this exam relate to the scenario on the last page. You may detach and recycle the last page if you like.

(a) [5 points] Part of the training data for the GooseDetection model comes from crowdsourced data collection through an app. Each submitted observation includes a photo (2-5 MB), location coordinates, timestamp, species identification, flock count, and behavior notes (text). Crowdsourced data often has quality problems, so batches of data labels are redundantly checked by trained volunteers (e.g., correcting species identification from “Canada Goose” to “Domestic Goose” or adjusting flock count). The photo itself is never changed, only the metadata labels. Approximately 15-20% of submitted observations have their metadata corrected once, and about 5% are corrected a second time.

Which versioning strategy is the most *space-efficient* for the **observations dataset** in this scenario (select one):

- ☐ Keeping copies of the dataset
- ☐ Tracking deltas
- ☐ Append-only data of observations (full records) with tracked offsets

Explain your answer, considering the size of records and frequency of modifications:

(b) [3 points] The fine tuning job for the GooseDetection model runs on a Kubernetes cluster to distribute the load. Exploring Kubernetes features, you wonder whether lifecycle hooks might be useful. What could you use a `preStop` lifecycle hook for in this setting (select all that apply; no justification required):

- ☐ To gracefully shut down a training pod by saving model checkpoints or uploading the latest training artifacts before the container is terminated
- ☐ To prevent Kubernetes from terminating the pod until the model finishes training
- ☐ To automatically scale the GooseDetection training job to more replicas when GPU utilization gets high
- ☐ To inject environment variables into the training container right before it starts executing its main process
- ☐ None of the above

(c) [5 points] After a community complaint about excessively noisy interventions to relocate geese from Hazelwood, you investigate which predictions of the *ResettlementPlanner* triggered those interventions and found that the used *FlightPredictor* model is probably problematic. Hence you next want to trace back (a) which version of the *FlightPredictor* model was used, (b) which data that model was trained on, (c) which *GooseDetection* model version was used to create part of that data, and (d) what data was used to train that *GooseDetection* model.

You recently introduced your team to *DVC* (Data Version Control) to track training datasets and model files in your team. Of the four pieces of information to track mentioned above, explain what DVC can help track and what additional information needs to be tracked outside of DVC for a complete investigation:

What DVC tracks:

What needs to be tracked outside DVC:

(c) [5 points] The *GooseDetection* model is based on a 500GB foundation model. It is very large and takes significant time to load into memory and deploy. Deploying and updating such a large model can be tricky, and the amount of MLOps tooling available can be overwhelming. Provide an example of **prudent and deliberate technical debt** related to deploying or scaling the machine learning model that would be plausible in the GooseGuard scenario. Your answer must explain why it is prudent and why it is deliberate.

Example:

Why prudent:

Why deliberate:

Question 2: Fairness [11 points]

During offline evaluation, you discover a concerning pattern: Due to variations in camera quality and training data distribution, the *GooseDetection* model's accuracy differs significantly between city neighborhoods.

(a) [4 points] There is a risk of deploying more frequent relocation interventions (e.g., acoustic devices) in low-income neighborhoods than in high-income neighborhoods, due to the *GooseDetection* model falsely detecting more geese in areas with lower quality cameras. This can create:

☐ a harm of allocation ☐ a harm of representation ☐ both ☐ neither

Brief explanation:

(b) [3 points] You have access to the last 10,000 images analyzed with the *GooseDetection* model: 6,000 for low-income neighborhoods and 4,000 for high-income neighborhoods. For each, you can see the number of geese the model detected. However, you do not have reliable labels about how many geese there actually are in those neighborhoods or in those specific pictures. Is there any notion of fairness that you could measure with this information you have now? (select all that apply; no explanation required)

☐ anti-classification ☐ group fairness ☐ equalized odds ☐ neither

(c) [4 points] A community representative is concerned about the fairness issue and insists that the *GooseDetection* model should only use the pictures but not any additional information about the zip code, income levels, or neighborhood name to make the model fair. Do you think this approach will effectively address the fairness concerns? Briefly explain why or why not?

Question 3: Explainability/Transparency [10 points]

(a) [5 points] The development team wants to understand the *global model behavior* of the *FlightPredictor* model (i.e., what patterns the model has learned overall, rather than why it made a specific prediction). Name a technique for understanding global model behavior that your team could use and explain briefly how the technique works.

Technique's name:

Basic idea of how the technique works (rough intuition is sufficient, 2-3 sentences):

(b) [5 points] The Airport Authority wants to publish a "Transparency Report" for community stakeholders explaining how *GooseGuard AI* works. A team member suggests: "Let's just publish all our model weights and architecture details on Hugging Face; that's maximum transparency!"

Is this approach likely to achieve meaningful transparency for community stakeholders (select all that apply)?

- ☐ Yes, providing all technical details is the best form of transparency
- ☐ No, most stakeholders cannot interpret model weights; meaningful transparency requires accessible explanations
- ☐ Yes, but only if we also provide training data
- ☐ Yes, releasing weights ensures all community members can verify model safety
- ☐ No, we should provide no explanations to protect intellectual property

Brief justification:

Question 4: Safety and Security [11 points]

(a) [6 points] Consider a potential *targeted poisoning attack* where an adversary (e.g., industrial saboteur, extremist group, or state actor) attempts to compromise the system's ability to protect aircraft at the PIT airport. Answer the four prompts below. Your answer must be reasonably realistic in the scenario and demonstrate an understanding of targeted poisoning attacks.

Attacker goal (intended outcomes):

Which security property does the attack most directly intend to undermine (select one):

☐ Confidentiality ☐ Integrity ☐ Availability

Attack method (what the attacker would do):

Mitigation strategy (how the developers can prevent the attack/make it harder):

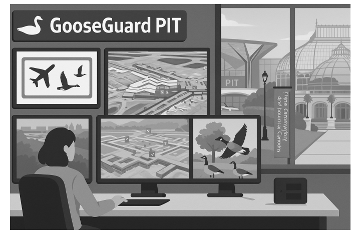
(c) [5 points] A data scientist on your team claims the *GooseDetection* model they fine-tuned is "very robust" and provides the following evidence: "We measured prediction stability on the test dataset when adding 10% random noise to each pixel of the test images. The predicted number of geese changed by less than 5% per image on average."

This robustness measure is relevant to which purposes of robustness (select all that apply)?

☐ Handling natural variations in input data (e.g., weather conditions, hardware problems)
☐ Defending against adversarial attacks (maliciously crafted inputs)
☐ Handling natural distribution shift (temporal robustness)
☐ Ensuring the model works on out-of-distribution data
☐ None of the above

Briefly explain your answer:

Scenario: Migratory Wildlife Management and Safety System



You are a team member at a consulting company with a contract from the City of Pittsburgh and the Allegheny County Airport Authority (PIT) to turn **GooseGuard AI** from a research prototype at the CMU Robotics Institute into a deployed project that positively improves environmental safety in the city and at the airport. The prototype was built by graduate students who are strong AI researchers but lacked software engineering experience. Research papers have been prioritized over robust infrastructure.

The GooseGuard AI mission is to use cutting-edge Machine Learning and vision systems to manage the interaction between dense urban wildlife and critical infrastructure, specifically focusing on **Canada geese** populations like those that inhabit **Schenley Park**, especially as it comes to aviation infrastructure around the **PIT Airport**. Collisions between birds (especially larger birds like Canada geese) and airplanes are known as *bird strikes*. Bird strikes can be dangerous and have led to many aviation accidents in the past, including the famous 2009 'Miracle on the Hudson' emergency landing in New York.

The timing is critical: PIT's new airport terminal opened in November 2025, significantly altering the landscape adjacent to flight paths. With 10 million travelers annually, any wildlife incidents pose major safety risks. The goal is to maximize safety by reducing bird strike risks while optimizing ethical wildlife management.

The core of the system consists of three models that build on each other:

GooseDetection: The model detects geese in photos, both on the ground and in the air. It is a fine-tuned version of a vision foundation model for wildlife detection. This model is used for counting geese populations and for detecting flying swarms. The initial training dataset included publicly available sources of wildlife pictures and video sources from the city (traffic cameras), as well as community-contributed observations (photos and metadata submitted through an app). For example, students in our class regularly see geese in Schenley Park outside the classroom windows and have contributed labeled photos and behavior annotations to the project, making CMU's campus area one of the best-represented regions in the training data. Beyond that, camera infrastructure varies dramatically across monitored areas. Affluent neighborhoods like Squirrel Hill and Point Breeze have high-resolution 4K cameras (installed by homeowner associations) with 98% uptime. The CMU campus and new PIT terminal have state-of-the-art cameras, with the campus area further enriched by student observations and annotations. Lower-income areas like Hazelwood and parts of Homewood rely on older municipal traffic cameras with 720p resolution and frequent outages. This disparity has resulted in training data heavily concentrated in affluent neighborhoods, the campus, and the terminal area, while lower-income residential areas are underrepresented.

FlightPredictor: This model predicts the probability of a swarm of geese flying in an area in the next 10 minutes. It is a deep learning model for tabular data, mostly time series geese populations in areas created with the *GooseDetection* model, past observed flights, and weather data. This is used to create alerts at the airport when the model predicts a high risk of bird strikes.

ResettlementPlanner: This optimization algorithm uses the *FlightPredictor* model to answer what-if questions about what would happen if geese populations were moved. This is used to create recommendations for *relocation interventions* that move geese away from critical areas. These relocation interventions include acoustic deterrents (loud broadcast devices), habitat modification, and other programs that encourage geese to relocate to designated “safe zones” in other areas. The *ResettlementPlanner* component is used to automatically dispatch wildlife management teams based on predicted risk levels and predicted risk reduction opportunities.

You work for a large consulting firm (like Accenture or PwC), where you are part of a small team that has the contract to transform this research prototype into a scalable, secure, and fair product. Your team has mostly junior team members proficient in data science and you are the team member with the most engineering experience. However, you have access to resources of the entire large consulting company.